Web Programming

Lab Assignment 13

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LAB 13 – HTML Code:

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Canvas Grid Drawing</title>
    <link rel="stylesheet" href="styles.css">
</head>
<body>
    <h1>Canvas Drawing Grid</h1>
    <div class="canvas-container">
        <div class="flip-card">
            <div class="flip-card-inner">
                <div class="flip-card-front">
                    <canvas id="canvas1"></canvas>
                </div>
                <div class="flip-card-back">
                    <div class="label">Mickey Mouse</div>
                </div>
            </div>
        </div>
        <div class="flip-card">
            <div class="flip-card-inner">
                <div class="flip-card-front">
                    <canvas id="canvas2"></canvas>
                </div>
                <div class="flip-card-back">
                    <div class="label">Smiley</div>
                </div>
            </div>
        </div>
        <div class="flip-card">
            <div class="flip-card-inner">
                <div class="flip-card-front">
                    <canvas id="canvas3"></canvas>
                </div>
                <div class="flip-card-back">
                    <div class="label">House</div>
                </div>
            </div>
        </div>
        <div class="flip-card">
```

```html
            <div class="flip-card-inner">
                <div class="flip-card-front">
                    <canvas id="canvas4"></canvas>
                </div>
                <div class="flip-card-back">
                    <div class="label">Atom</div>
                </div>
            </div>
        </div>
    </div>
    <script src="script.js"></script>
</body>
</html>
```

LAB 13 – CSS Code:

```
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

body {
    display: flex;
    align-items: center;
    justify-content: center;
    background: linear-gradient(135deg, #1a1a2e 0%, #16213e 100%);
    color: #e0e0e0;
    font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
    min-height: 100vh;
    padding: 40px 20px;
}

h1 {
    text-align: center;
    margin-bottom: 40px;
    font-size: 2.5rem;
    background: linear-gradient(135deg, #00d4ff, #0099ff);
    -webkit-background-clip: text;
    -webkit-text-fill-color: transparent;
    background-clip: text;
    text-shadow: 0 4px 15px rgba(0, 212, 255, 0.3);
}

.canvas-container {
    width: min(90vw, 90vh);
    aspect-ratio: 1;
    display: grid;
    grid-template-columns: repeat(2, 1fr);
    grid-template-rows: repeat(2, 1fr);
    gap: 30px;
    margin: 0 auto;
    padding: 20px;
}

canvas {
    background: linear-gradient(135deg, #0f3460 0%, #1a5276 100%);
    border: 2px solid #00d4ff;
```

```
    border-radius: 12px;
    box-shadow: 0 8px 32px rgba(0, 212, 255, 0.2),
                inset 0 0 20px rgba(0, 212, 255, 0.1);
    transition: all 0.3s ease;
    cursor: pointer;

    width: 100%;
    height: 100%;
    aspect-ratio: 1;
}

canvas:hover {
    transform: translateY(-5px);
    box-shadow: 0 12px 40px rgba(0, 212, 255, 0.4),
                inset 0 0 20px rgba(0, 212, 255, 0.2);
    border-color: #00ffff;
}

.flip-card {
    background-color: transparent;
    perspective: 1000px;
    width: 100%;
    height: 100%;
}

.flip-card-inner {
    position: relative;
    width: 100%;
    height: 100%;
    text-align: center;
    transition: transform 0.8s;
    transform-style: preserve-3d;
}

.flip-card.flipped .flip-card-inner {
    transform: rotateY(180deg);
}

.flip-card-front, .flip-card-back {
    position: absolute;
    width: 100%;
    height: 100%;
    -webkit-backface-visibility: hidden;
    backface-visibility: hidden;
    border-radius: 12px;
}

.flip-card-front {
    display: flex;
    align-items: center;
```

```
    justify-content: center;
}

.flip-card-back {
    transform: rotateY(180deg);
    background: linear-gradient(135deg, #0f3460 0%, #1a5276 100%);
    border: 2px solid #00d4ff;
    border-radius: 12px;
    box-shadow: 0 8px 32px rgba(0, 212, 255, 0.2),
                inset 0 0 20px rgba(0, 212, 255, 0.1);
    display: flex;
    align-items: center;
    justify-content: center;
}

.label {
    font-size: 1.5rem;
    font-weight: bold;
    color: #00d4ff;
    text-shadow: 0 2px 10px rgba(0, 212, 255, 0.5);
}
```

LAB 13 – JavaScript Code:

```
let canvas1, canvas2, canvas3, canvas4;

function resizeCanvases() {
    [canvas1, canvas2, canvas3, canvas4].forEach(c => {
        c.width = c.clientWidth;
        c.height = c.clientHeight;
    });
    drawAll();
}

function drawAll() {
    drawMickey(canvas1.getContext('2d'));
    drawSmiley(canvas2.getContext('2d'));
    drawHouse(canvas3.getContext('2d'));
    drawAtom(canvas4.getContext('2d'));
}

function drawMickey(ctx) {
    const w = ctx.canvas.width;
    ctx.clearRect(0, 0, w, w);
    const scale = w / 300;
    ctx.save();
    ctx.scale(scale, scale);

    // Ears (black)
    ctx.fillStyle = '#000';
    ctx.beginPath();
    ctx.arc(80, 80, 35, 0, Math.PI * 2);
    ctx.fill();
    ctx.beginPath();
    ctx.arc(220, 80, 35, 0, Math.PI * 2);
    ctx.fill();

    // Face (light skin tone)
    ctx.fillStyle = '#fde3cb';
    ctx.beginPath();
    ctx.arc(150, 150, 80, 0, Math.PI * 2);
    ctx.fill();
    ctx.strokeStyle = '#000';
    ctx.lineWidth = 3;
    ctx.stroke();
```

```
    // Eyes (two black ovals with black outlines)
    ctx.fillStyle = '#000';
    ctx.beginPath();
    ctx.ellipse(128, 120, 10, 16, 0, 0, Math.PI * 2);
    ctx.fill();
    ctx.beginPath();
    ctx.ellipse(172, 120, 10, 16, 0, 0, Math.PI * 2);
    ctx.fill();

    // Nose (black)
    ctx.fillStyle = '#000';
    ctx.beginPath();
    ctx.ellipse(150, 157, 10, 8, 0, 0, Math.PI * 2);
    ctx.fill();

    // Mouth (black curve with a line underneath to create a smile; filled with white to
create an open mouth effect)
    ctx.strokeStyle = '#000';
    ctx.lineWidth = 5;
    ctx.beginPath();
    ctx.arc(150, 180, 30, 0, Math.PI);
    ctx.stroke();
    ctx.beginPath();
    ctx.lineTo(117, 180);
    ctx.lineTo(183, 180);
    ctx.stroke();
    ctx.fillStyle = '#fff';
    ctx.beginPath();
    ctx.arc(150, 180, 30, 0, Math.PI);
    ctx.fill();
    ctx.restore();
}

function drawSmiley(ctx) {
    const w = ctx.canvas.width;
    const scale = w / 300;
    ctx.clearRect(0,0,w,w);
    ctx.save();
    ctx.scale(scale, scale);
    ctx.fillStyle = '#FFD700';
    ctx.beginPath();
    ctx.arc(150, 150, 80, 0, Math.PI * 2);
    ctx.fill();
    ctx.strokeStyle = '#000';
    ctx.lineWidth = 3;
    ctx.stroke();
    ctx.fillStyle = '#000';
    ctx.beginPath();
    ctx.arc(122, 125, 12, 0, Math.PI * 2);
    ctx.fill();
```

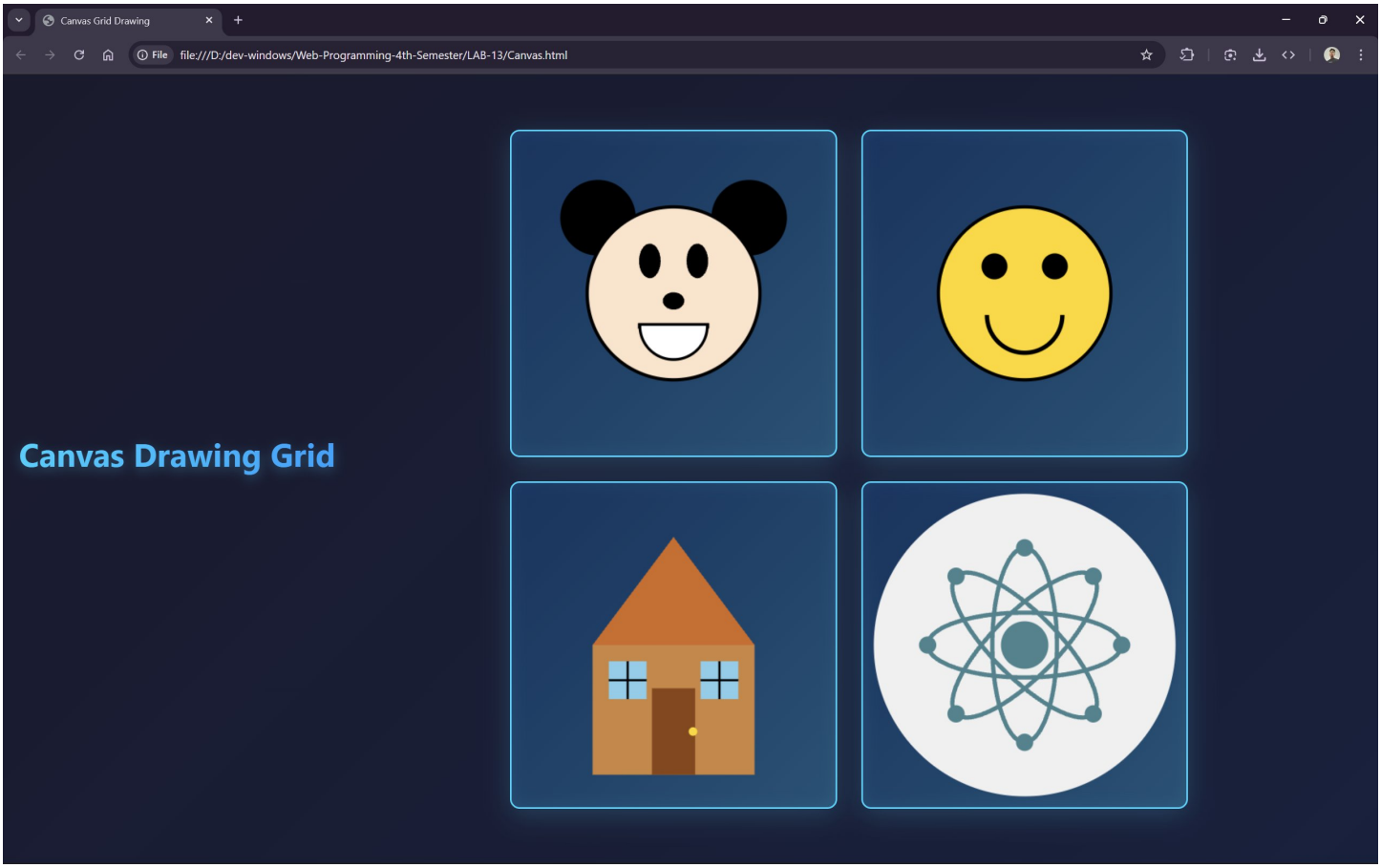
```
    ctx.beginPath();
    ctx.arc(178, 125, 12, 0, Math.PI * 2);
    ctx.fill();
    ctx.strokeStyle = '#000';
    ctx.lineWidth = 4;
    ctx.beginPath();
    ctx.arc(150, 170, 35, 0, Math.PI);
    ctx.stroke();
    ctx.restore();
}

function drawHouse(ctx) {
    const w = ctx.canvas.width;
    const scale = w / 300;
    ctx.clearRect(0,0,w,w);
    ctx.save();
    ctx.scale(scale, scale);
    ctx.fillStyle = '#D2691E';
    ctx.beginPath();
    ctx.moveTo(75, 150);
    ctx.lineTo(225, 150);
    ctx.lineTo(150, 50);
    ctx.closePath();
    ctx.fill();
    ctx.fillStyle = '#CD853F';
    ctx.fillRect(75, 150, 150, 120);
    ctx.fillStyle = '#8B4513';
    ctx.fillRect(130, 190, 40, 80);
    ctx.fillStyle = '#FFD700';
    ctx.beginPath();
    ctx.arc(168, 230, 4, 0, Math.PI * 2);
    ctx.fill();
    ctx.fillStyle = '#87CEEB';
    ctx.fillRect(90, 165, 35, 35);
    ctx.fillRect(175, 165, 35, 35);
    ctx.strokeStyle = '#000';
    ctx.lineWidth = 2;
    ctx.beginPath();
    ctx.moveTo(107.5, 165);
    ctx.lineTo(107.5, 200);
    ctx.stroke();
    ctx.beginPath();
    ctx.moveTo(90, 182.5);
    ctx.lineTo(125, 182.5);
    ctx.stroke();
    ctx.beginPath();
    ctx.moveTo(192.5, 165);
    ctx.lineTo(192.5, 200);
    ctx.stroke();
    ctx.beginPath();
```

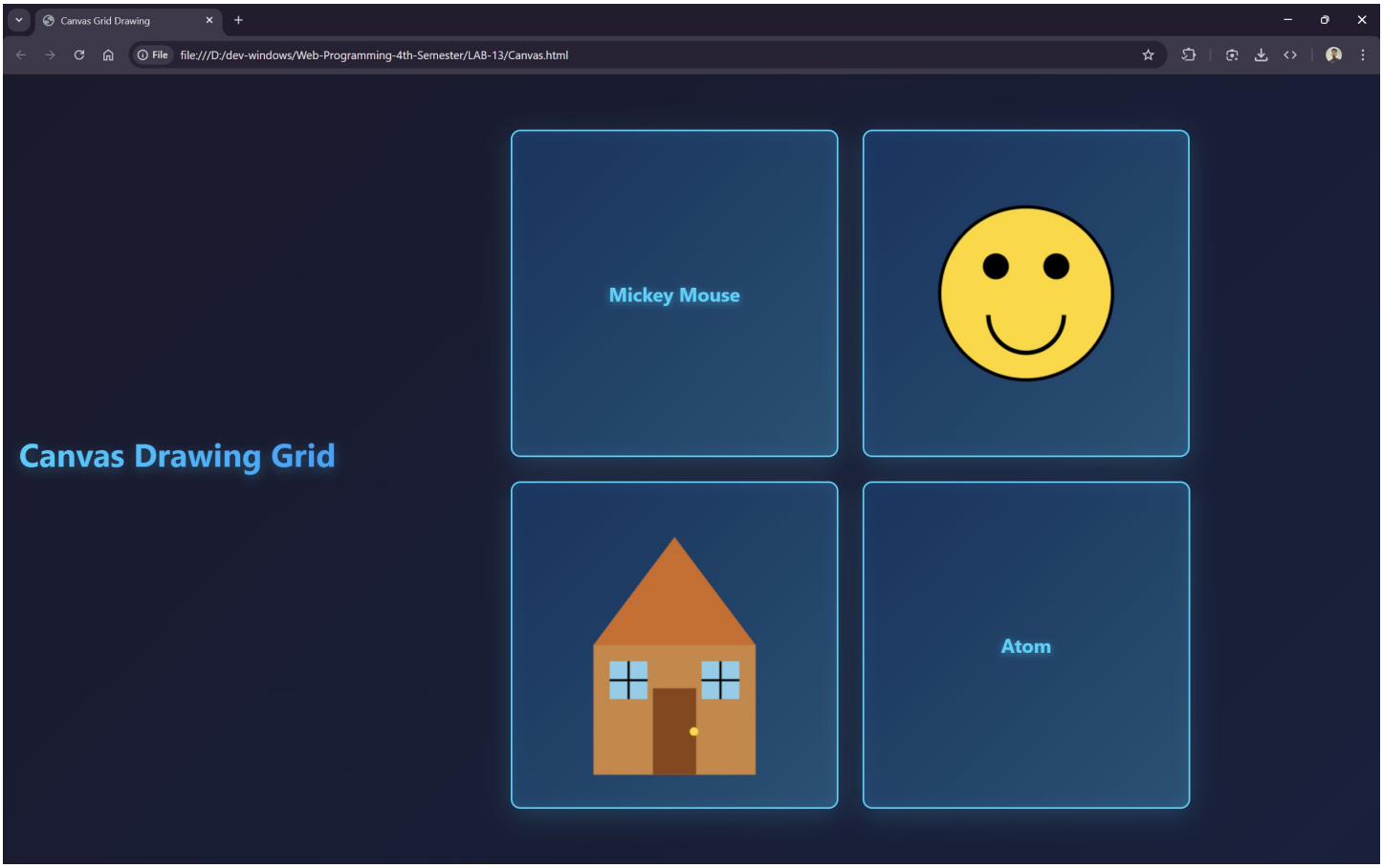
```
    ctx.moveTo(175, 182.5);
    ctx.lineTo(210, 182.5);
    ctx.stroke();
    ctx.restore();
}

function drawAtom(ctx) {
    const w = ctx.canvas.width;
    const scale = w / 300;
    ctx.clearRect(0, 0, w, w);
    ctx.save();
    ctx.scale(scale, scale);

    // Background circle (light gray)
    ctx.fillStyle = '#f0f0f0';
    ctx.beginPath();
    ctx.arc(150, 150, 140, 0, Math.PI * 2);
    ctx.fill();

    // Electron orbits (blue)
    ctx.strokeStyle = '#47848F';
    ctx.lineWidth = 4;

    // Horizontal orbit
    ctx.beginPath();
    ctx.ellipse(150, 150, 90, 30, 0, 0, Math.PI * 2);
    ctx.stroke();

    // Vertical orbit
    ctx.beginPath();
    ctx.ellipse(150, 150, 30, 90, 0, 0, Math.PI * 2);
    ctx.stroke();

    // Diagonal orbit (45 degrees)
    ctx.beginPath();
    ctx.ellipse(150, 150, 90, 30, Math.PI / 4, 0, Math.PI * 2);
    ctx.stroke();

    // Diagonal orbit (135 degrees)
    ctx.beginPath();
    ctx.ellipse(150, 150, 30, 90, Math.PI / 4, 0, Math.PI * 2);
    ctx.stroke();

    // Nucleus (blue)
    ctx.fillStyle = '#47848F';
    ctx.beginPath();
    ctx.arc(150, 150, 22, 0, Math.PI * 2);
    ctx.fill();

    // Electrons (blue)
```

```
    ctx.fillStyle = '#47848F';

    // Top electron
    ctx.beginPath();
    ctx.arc(150, 60, 8, 0, Math.PI * 2);
    ctx.fill();

    // Bottom electron
    ctx.beginPath();
    ctx.arc(150, 240, 8, 0, Math.PI * 2);
    ctx.fill();

    // Left electron
    ctx.beginPath();
    ctx.arc(60, 150, 8, 0, Math.PI * 2);
    ctx.fill();

    // Right electron
    ctx.beginPath();
    ctx.arc(240, 150, 8, 0, Math.PI * 2);
    ctx.fill();

    // Top-left electron
    ctx.beginPath();
    ctx.arc(86.36, 86.36, 8, 0, Math.PI * 2);
    ctx.fill();

    // Top-right electron
    ctx.beginPath();
    ctx.arc(213.64, 86.36, 8, 0, Math.PI * 2);
    ctx.fill();

    // Bottom-left electron
    ctx.beginPath();
    ctx.arc(86.36, 213.64, 8, 0, Math.PI * 2);
    ctx.fill();

    // Bottom-right electron
    ctx.beginPath();
    ctx.arc(213.64, 213.64, 8, 0, Math.PI * 2);
    ctx.fill();

    ctx.restore();
}

// initialization
window.addEventListener('DOMContentLoaded', () => {
    canvas1 = document.getElementById('canvas1');
    canvas2 = document.getElementById('canvas2');
    canvas3 = document.getElementById('canvas3');
```

```
    canvas4 = document.getElementById('canvas4');
    resizeCanvases();

    // Add flip functionality
    document.querySelectorAll('.flip-card').forEach(card => {
        card.addEventListener('click', () => {
            card.classList.toggle('flipped');
        });
    });
});

window.addEventListener('resize', resizeCanvases);
```

LAB 13 – Screenshots:




Canvas Grid Drawing
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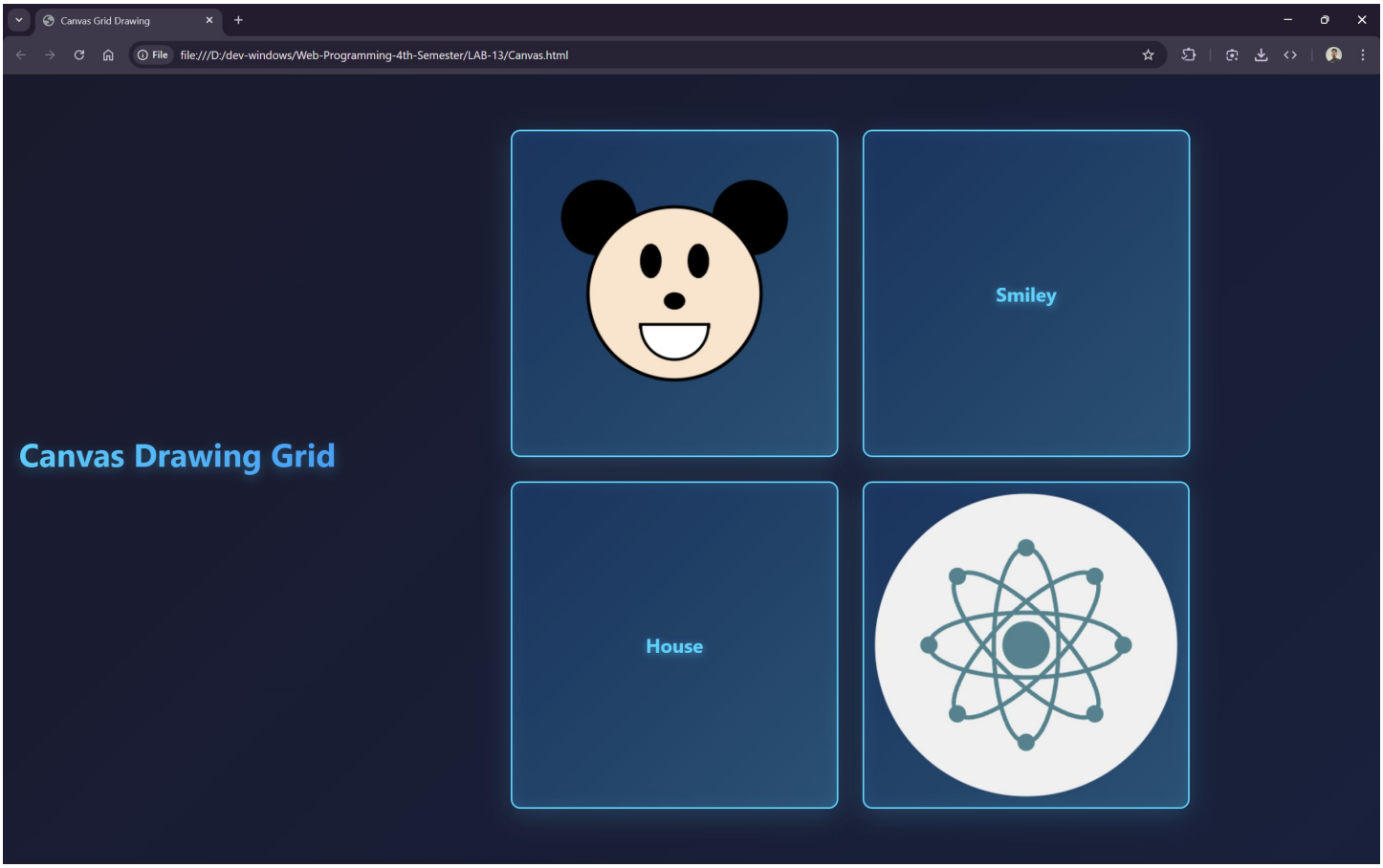
Canvas Drawing Grid
Mickey Mouse
Atom


Canvas Grid Drawing
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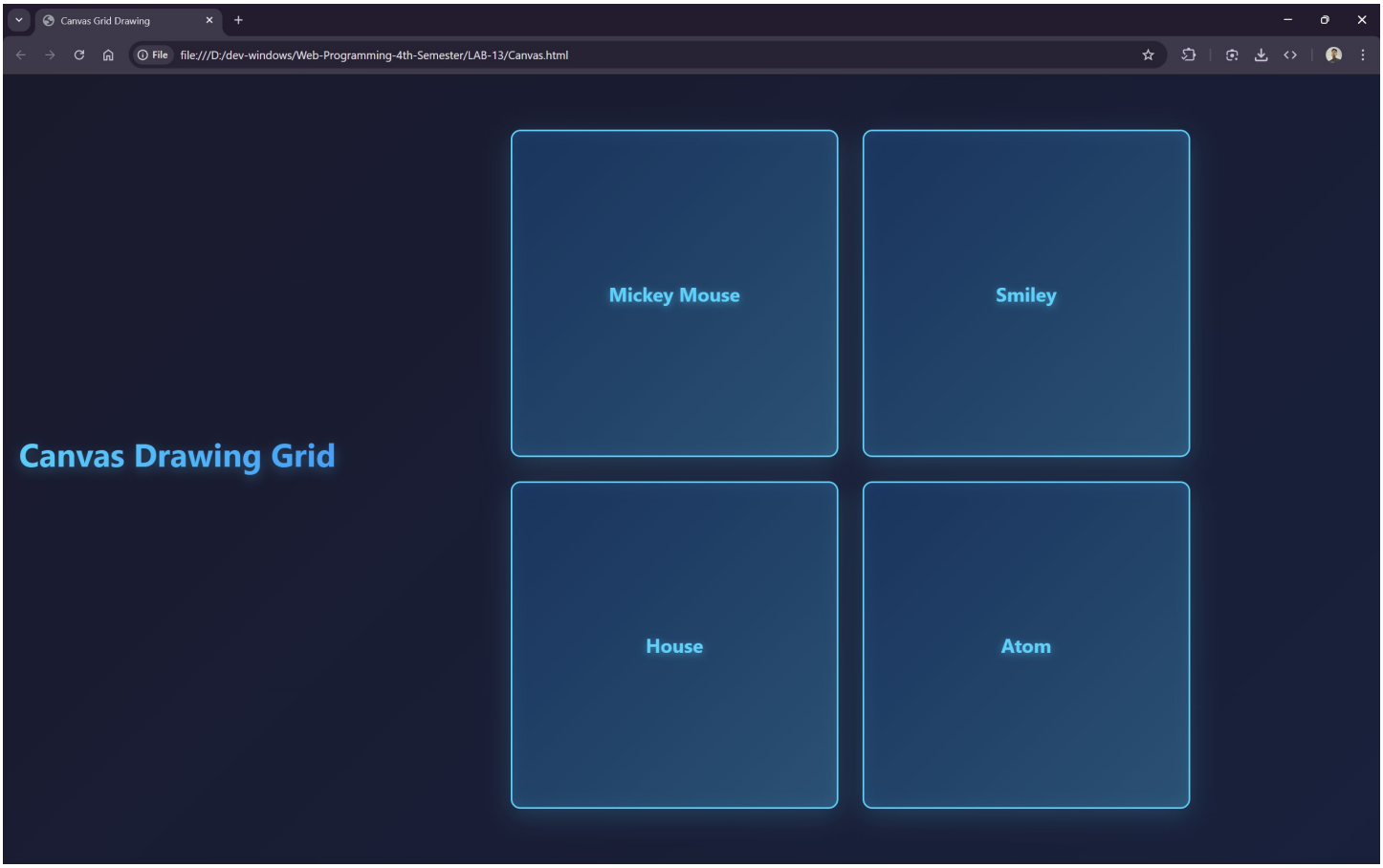
Canvas Drawing Grid
Smiley
House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Canvas Drawing Grid
Mickey Mouse
Smiley
House
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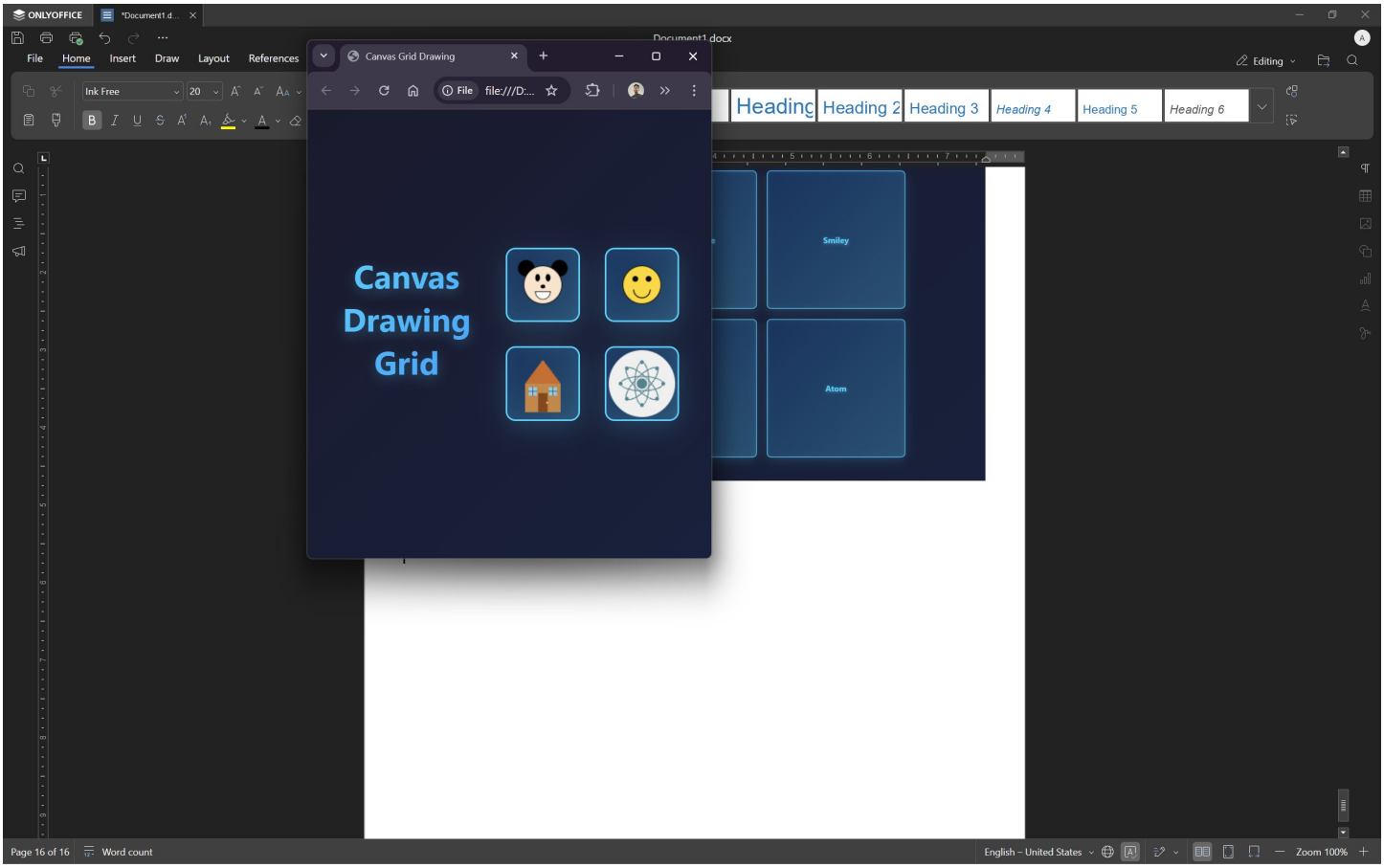
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Word count
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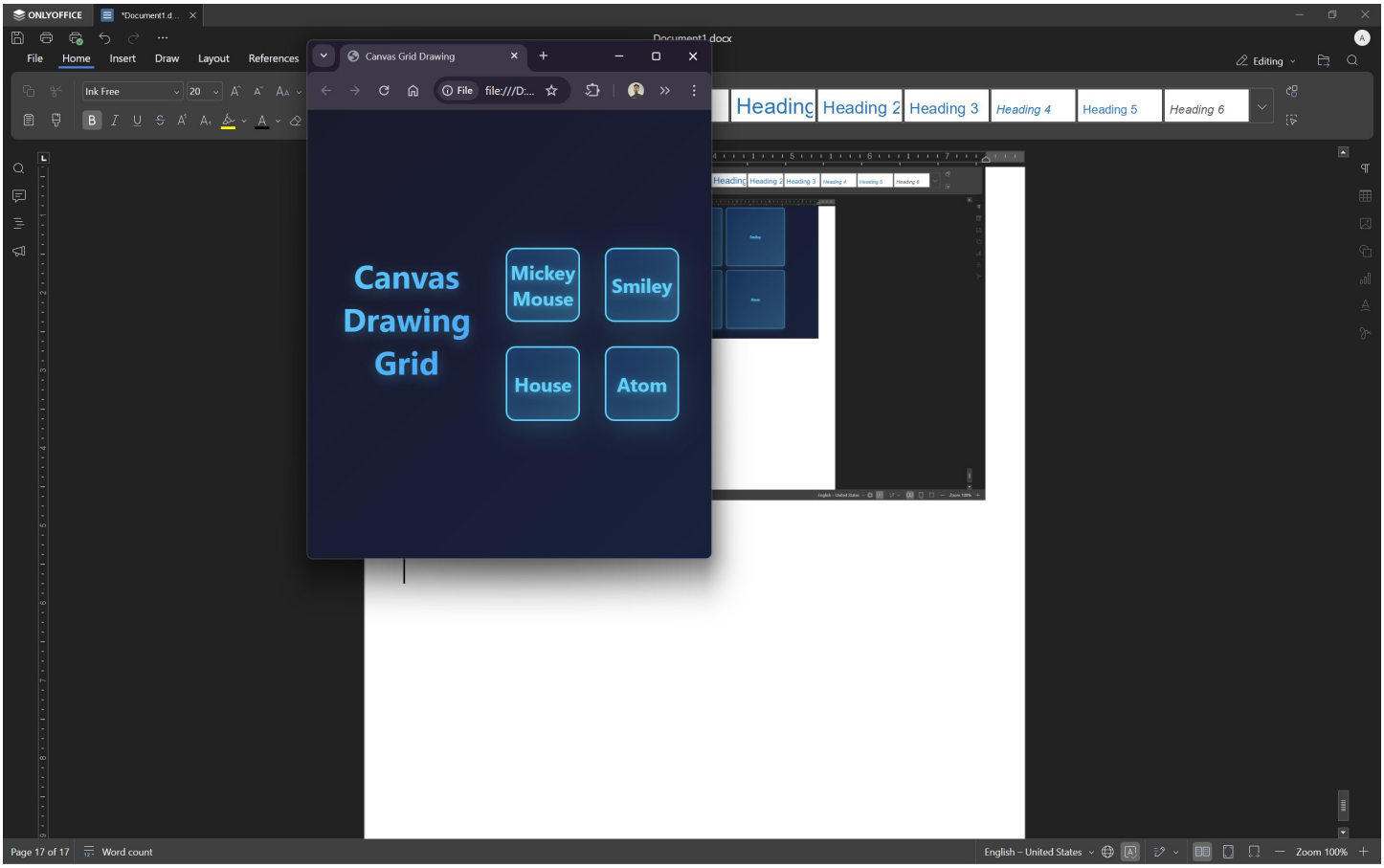
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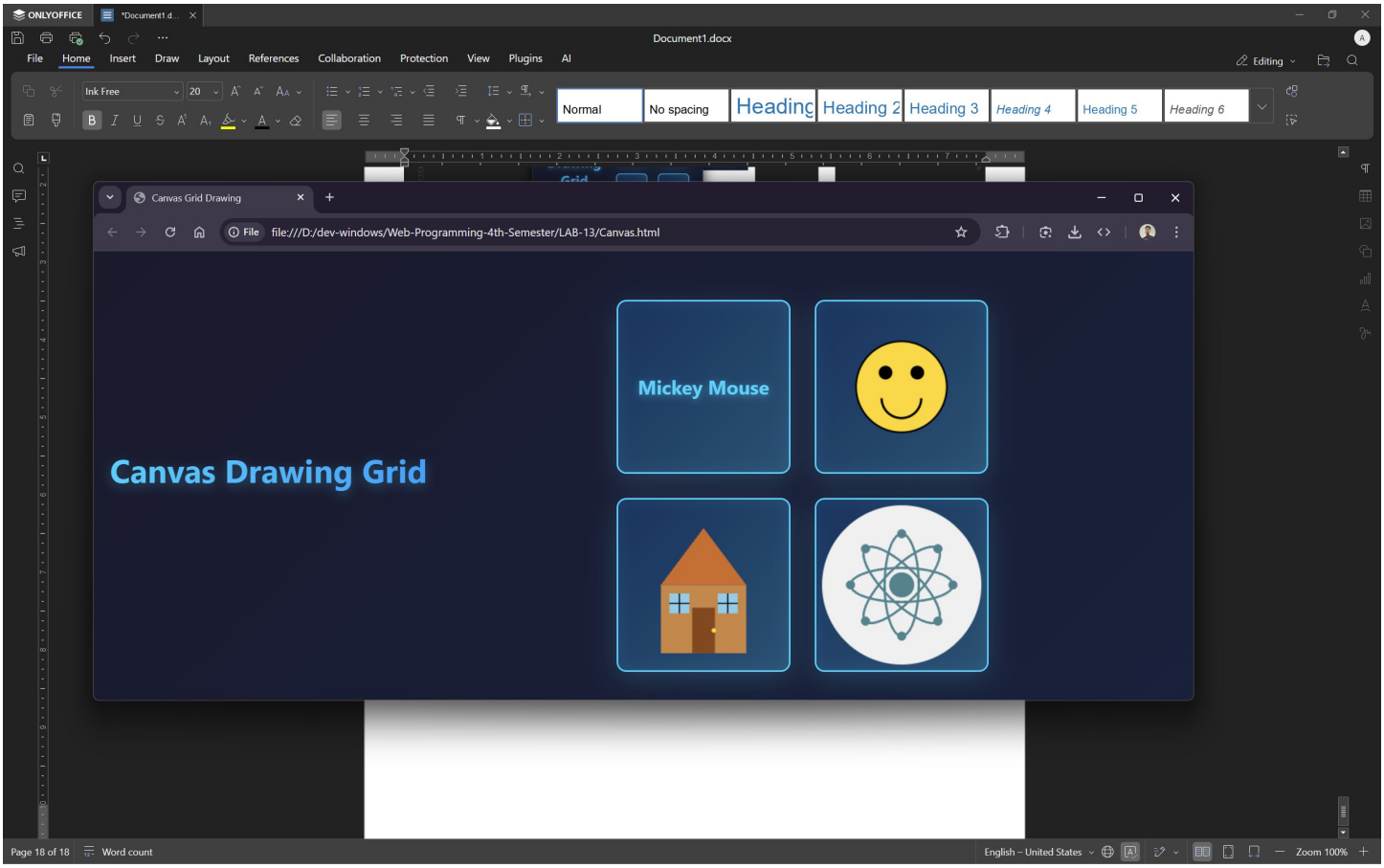
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Canvas Grid Drawing
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Canvas Drawing Grid
Mickey Mouse
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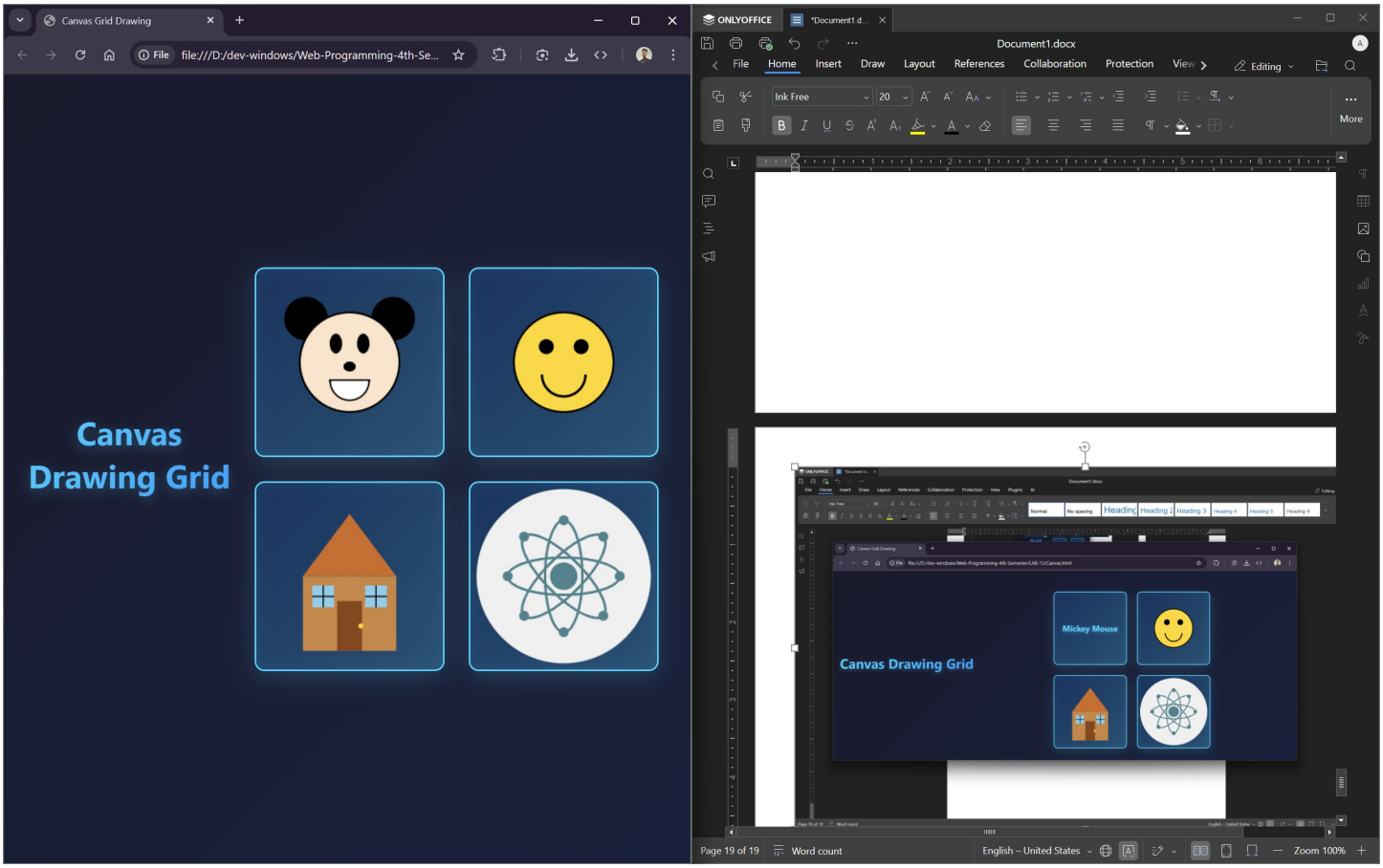
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THANK YOU!!!